INTERNSHIP TASK REPORT FOR SIEM (LOGPOINT)

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PREPARED BY

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INTERN

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Installing kali Linux in Virtual box

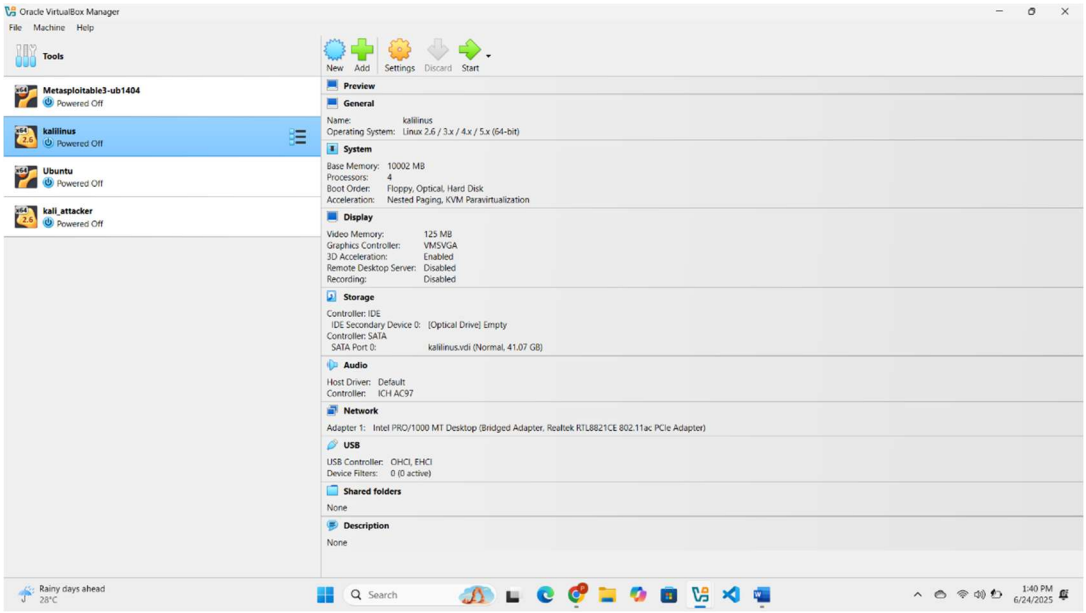
Kali Linux is a hacking-focused Linux distro used by ethical hackers and security pros to test, analyze, and secure systems. Kali is not a general-purpose OS like Windows or Ubuntu. It's a powerful toolkit intended for users with knowledge of networking, Linux, and cybersecurity.

Steps for setting up Kali Linux

1. **Install VirtualBox**
    - Download from: virtualbox.org
    - Choose your OS and install it.
2. **Download Kali Linux ISO**
    - Get the **Installer ISO (64-bit)** from: kali.org
3. **Create a Virtual Machine**
    - Open VirtualBox → Click **New**
    - Name: *Kali Linux*, Type: *Linux*, Version: *Debian (64-bit)*
4. **Assign Resources**
    - RAM: Minimum 2 GB (4 GB recommended)
    - Hard Disk: Create a 20 GB **dynamically allocated VDI**
5. **Mount Kali ISO**
    - Go to **Settings > Storage**
    - Load the downloaded ISO in the optical drive
6. **Network Configuration (Optional)**
    - Use **Bridged Adapter** or **NAT** under **Settings > Network**
7. **Install Kali Linux**
    - Start VM → Choose **Graphical Install**
    - Set language, location, user, and password
    - Use **guided disk partitioning**

8. **Complete Setup**
    - Finish installation
    - Remove ISO from virtual drive before reboot

After the installation and configuration of Kali Linux within the virtual box. The same Kali linux was clone. Currently, the virtual box has two kali linux. One of the them is attacker from where we are going to perform scanning. And another one is the victim.



The ip address of both the devices are:

- Kali_attacker : 192.168.0.122 (Attacker)
- Kalilinux: 192.168.0.92 (Victim)

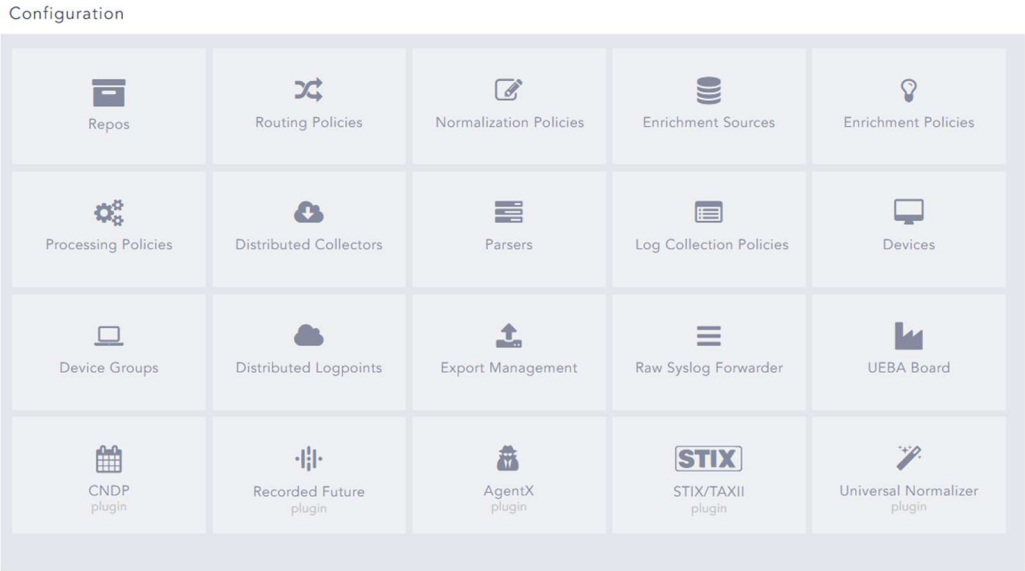
Forward Logs from Kali Linus to Logpoint

Introduction to Logpoint

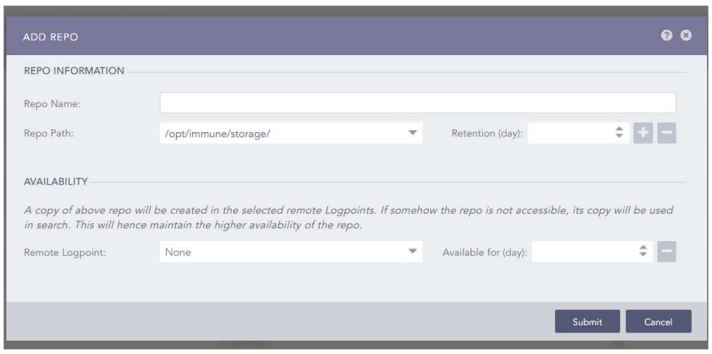
LogPoint is a Security Information and Event Management (SIEM) platform used to collect, analyze, and correlate log data from various sources like servers, firewalls, and applications. It helps organizations detect security threats, monitor user activity, investigate incidents, and meet compliance requirements. LogPoint uses a powerful query language called LEQL to normalize and analyze logs in real time, enabling security teams to create alerts, dashboards, and reports for proactive threat detection and response.

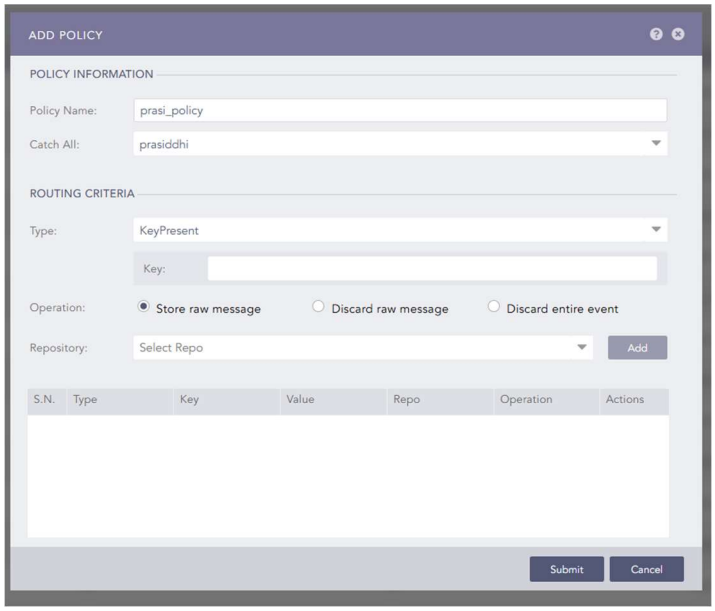
Logpoint setup:

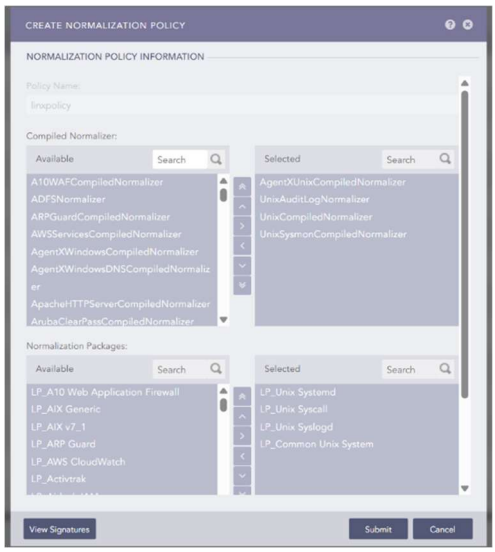
Before forwarding the logs to the logpoint. Different repo, routing policies, normalization policies, devices are created so the same kind of log are stored on the same repo and will be easier to access in the future.

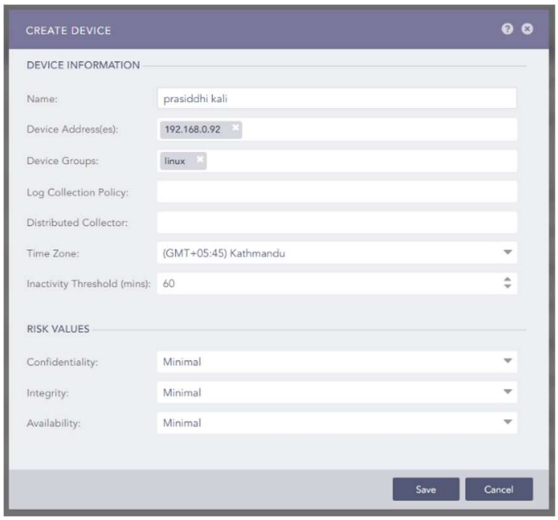


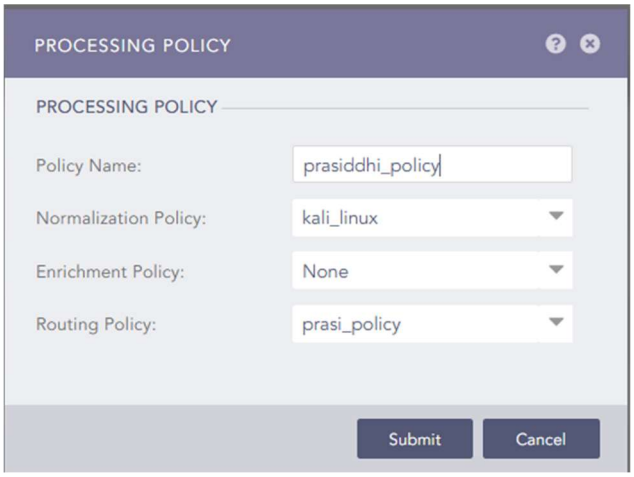
For adding new repo, routing policies, normalization policies, devices go to configuration and add the following.











To forward log to the logpoint. LogPoint uses **Syslog (UDP 514)** for log collection. You'll need to forward logs using **rsyslog**.

1. **Ensure rsyslog is installed and running**
   sudo apt update
   sudo apt install rsyslog
   sudo systemctl enable rsyslog
   sudo systemctl start rsyslog

2. **Edit rsyslog config**

   sudo nano /etc/rsyslog.conf

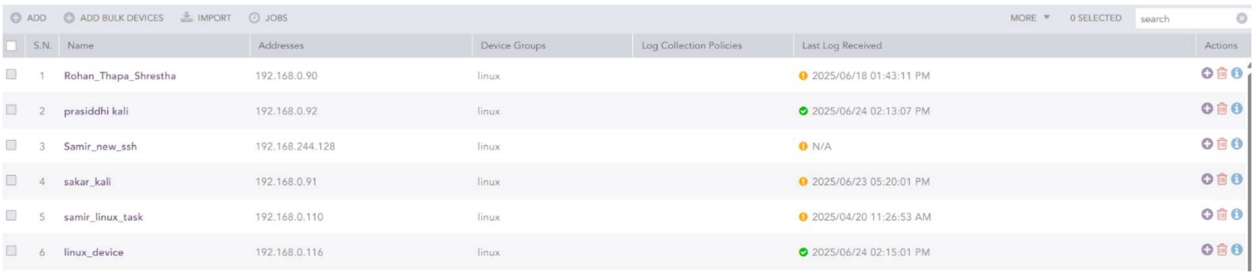
   Edit rsyslog.conf file **changes how your system logs are handled** including how they're stored, filtered, or **forwarded to external systems** like LogPoint.

```
ruleset(name="local2forwarder") {
    action(
        type="omfwd"
        target="192.168.0.9"
        port="514"
        protocol="tcp"
        action.resumeRetryCount="-1"
        queue.type="linkedList"
        queue.filename="lfqueue"
        queue.maxDiskSpace="1g"
        queue.saveOnShutdown="on"
        #template="localFormat"
    )
}
```

```
# Use default timestamp format
$ActionFileDefaultTemplate RSYSLOG_TraditionalFileFormat
$IncludeConfig /etc/rsyslog.d/*.conf
$WorkDirectory /var/spool/rsyslog
auth,authpriv.*    /var/log/auth.log
```

3. **Restart rsyslog**
   sudo systemctl restart rsyslog
   sudo systemctl status rsyslog

After completing the overall steps, the kalilinux device(victim) will be able to send the logs to the logpoint. If the kali attacker launch attacks like Nmap scan and any other scan to access the victim device the log will be forwarded to the logpoint.



Analyzing the Log.

2025/06/24 02:47:06 PM

log_ts=2025/06/24 02:47:06 PM | device_ip=192.168.0.92 | device_name=prasiddhi kali | col_type=syslog | source_address=192.168.0.92 | source_port=48522 | destination_address=192.168.0.9 | destination_port=514 | sig_id=19067 | repo_name=prasiddhi | severity=4 | facility=0 | protocol=TCP | process=kernel | URGP=0 | col_ts=2025/06/24 02:47:10 PM | collected_at=LogPoint | device_address=192.168.0.92 | index_ts=2025/06/24 02:47:10 PM | logpoint_name=LogPoint | norm_id=Kernel | outbound_interface=eth0 | packet_id=35314 | packet_length=254 | precedence=0x00 | reserved_bit=0x00 | time_to_live=64 | type_of_service=0x00 | window=502 |

<4>Jun 24 14:47:06 2400-1A00-B050-
ip6 kernel: [UFW AUDIT] IN= OUT=eth0 SRC=192.168.0.92 DST=192.168.0.9 LEN=254 TOS=0x00 PREC=0x00 TTL=64 ID=35314 DF PROTO=TCP SPT=48522 DPT=514 WINDOW=502 RES=0x00 ACK PSH URGP=0

Field	Value	Description
Timestamp	2025/06/24 14:40:20	Time the log was generated.
Source IP	192.168.0.92	Kali Defender machine.
Destination IP	192.168.0.9	LogPoint SIEM server.
Source Port	48522	Ephemeral port used to initiate the TCP connection.
Destination Port	514	Standard syslog port.
UFW Status	AUDIT	Logged by UFW; connection allowed and recorded.
Severity	4 (Warning)	Syslog severity level indicating a warning event.

This log entry shows that logs were successfully forwarded from the Kali Defender system to the LogPoint SIEM. The UFW firewall accepted the connection, and the message was relayed via TCP port 514, the normal syslog service port. This confirms that.

- The rsyslog setup is correct.
- LogPoint is getting Syslog notifications.
- The Defender's firewall configured to log audit-level events.

This type of log entry is required to verify the visibility and traceability of events between monitored hosts and the SIEM.

Setting up Alert Rules.

1. Authentication Failure

This rule detects brute-force or suspicious login behavior by identifying when a user experiences more than 10 failed SSH password login attempts from the same source address within a given time window (e.g., 1 minute). It helps security teams respond to potential attacks or unauthorized access attempts quickly.

* Query

```
norm_id=Unix "process"=sshd-sessions action=Failed object=password
| chart count() as failure_count by device_ip,user, source_address
| search failure_count > 10
```

Browse

Expand

Search

* Repos

prasiddhi ×

* Query Time-range

2 Minutes

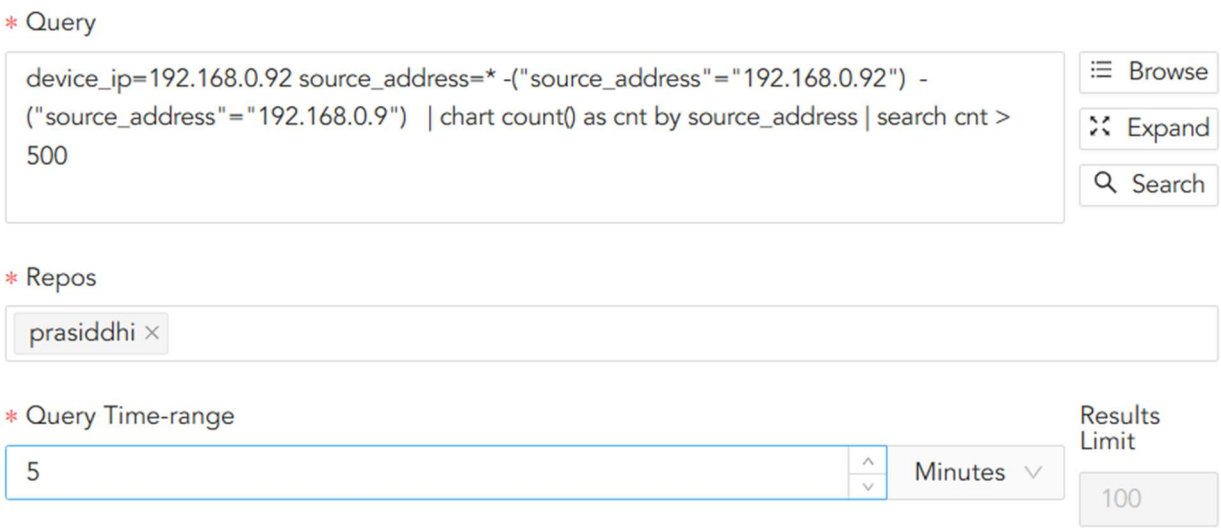
Results Limit

100

Search Interval

2. High Number of Requests Alert

This LogPoint alert rule monitors activity on the host device with IP `192.168.0.92` and identifies any external source IPs (excluding trusted IPs `192.168.0.92` and `192.168.0.9`) that generate more than 500 log events within a defined time window. It filters logs to include only those with a source address, counts the number of events per source IP, and triggers an alert if any single IP exceeds the threshold. This helps detect abnormal or suspicious activity such as brute-force attacks, port scanning, or other high-volume traffic that could indicate a potential security threat.



Simulating Attack scenario.

Two kali linux machine among which one of them is the attacker and another is the victim was used. When attack was performed from the attacker machine to the victim, the victim send the log to the logpoint.

Attack 1:

Brute Force Attack:

A brute force attack is a method used to gain unauthorized access to systems, accounts, or encrypted data by systematically trying all possible combinations of passwords, keys, or inputs until the correct one is found.

What was actually done?

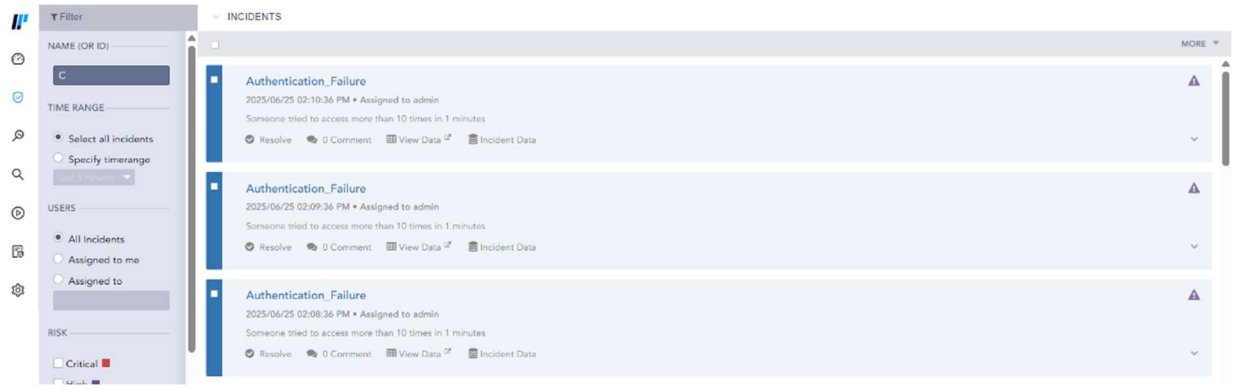
1. **Installed Hydra on Attacker Server**
   sudo apt update
   sudo apt install hydra

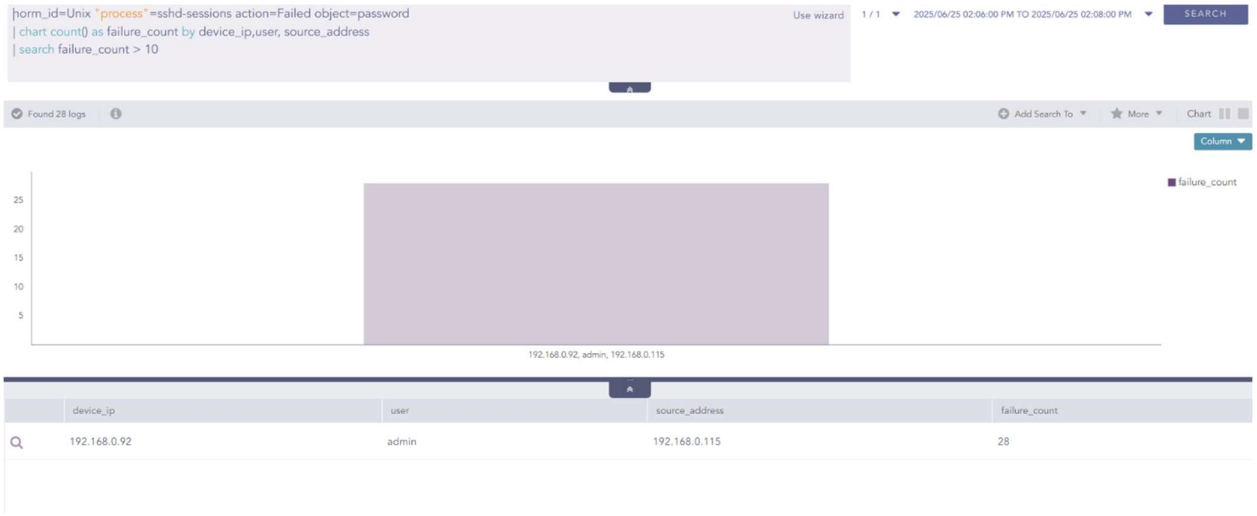
2. **Used preinstalled list in hydra**
   Located on: /usr/share/wordlists/rockyou.txt

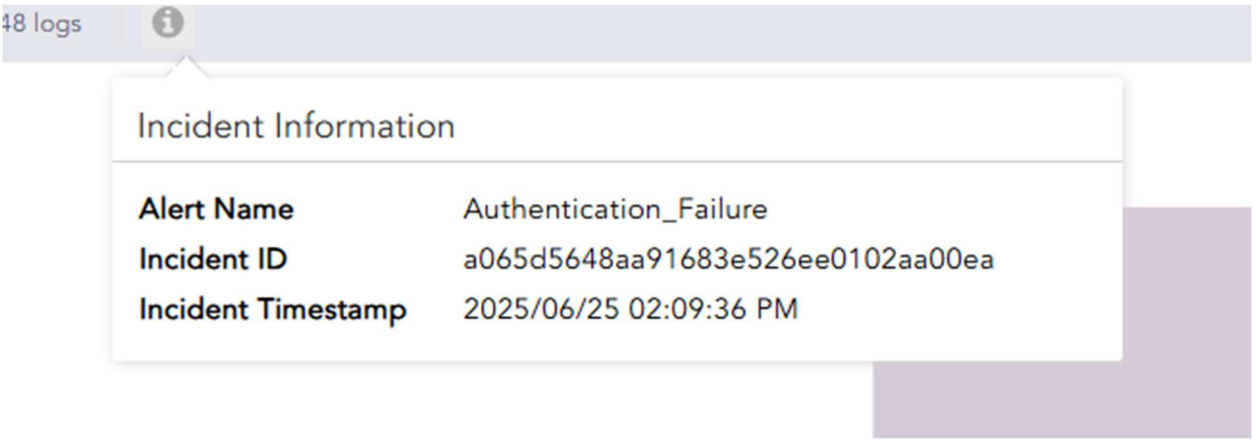
3. **Performed Brute Force using Hydra**
   hydra -l admin -P /usr/share/wordlists/rockyou.txt ssh://192.168.0.92

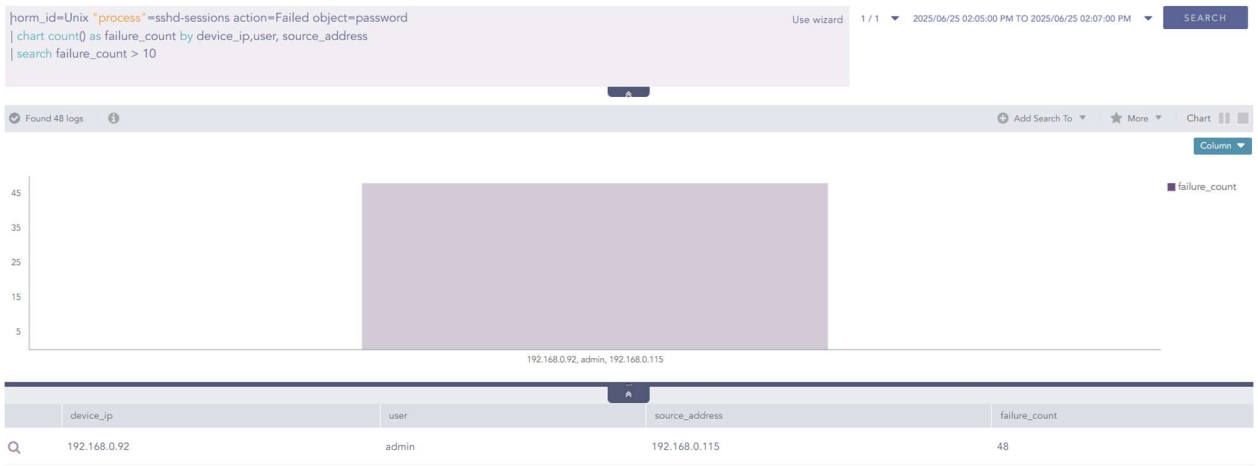
What actually happened?

Logpoint detected multiple failed FTP login attempts from various sources. These activities are indicative of a brute force attack, a method used by attackers to gain access by trying many username-password combinations. The Authentication_Failure alert rule was triggered.


Filter
INCIDENTS
NAME (OR ID)
C
TIME RANGE
Select all incidents
Specify timerange
USERS
All Incidents
Assigned to me
Assigned to
RISK
Critical
MORE
Authentication_Failure
2025/06/25 02:10:36 PM • Assigned to admin
Someone tried to access more than 10 times in 1 minutes
Resolve
0 Comment
View Data
Incident Data
Authentication_Failure
2025/06/25 02:09:36 PM • Assigned to admin
Someone tried to access more than 10 times in 1 minutes
Resolve
0 Comment
View Data
Incident Data
Authentication_Failure
2025/06/25 02:08:36 PM • Assigned to admin
Someone tried to access more than 10 times in 1 minutes
Resolve
0 Comment
View Data
Incident Data


horm_id=Unix "process"=sshd-sessions action=Failed object=password
| chart count() as failure_count by device_ip,user, source_address
| search failure_count > 10
Use wizard
1 / 1
2025/06/25 02:06:00 PM TO 2025/06/25 02:08:00 PM
SEARCH
Found 28 logs
Add Search To
More
Chart
Column
failure_count
25
20
15
10
5
192.168.0.92, admin, 192.168.0.115
device_ip
user
source_address
failure_count
192.168.0.92
admin
192.168.0.115
28


48 logs
Incident Information
Alert Name
Authentication_Failure
Incident ID
a065d5648aa91683e526ee0102aa00ea
Incident Timestamp
2025/06/25 02:09:36 PM



What it means:

The alert rule executed in Logpoint identified a suspicious login pattern indicating a potential brute-force attack. The system detected 28 failed SSH login attempts made by the user "admin" from the source IP address 192.168.0.115 targeting the device with IP 192.168.0.92. These repeated failed authentication attempts within a short time window suggest that an unauthorized user may be trying to gain access by guessing the password. Such activity typically warrants immediate attention, as it poses a security risk and may indicate the need for further investigation or mitigation actions, such as blocking the offending IP or enforcing stricter authentication controls.

Immediate Action:

1. Block the Attacker's IP Address: Use your firewall or system-level controls to block the malicious IP instantly.
2. Start Live Monitoring: Activate live alerts and monitoring in your SIEM (e.g..LogPoint) for:
    - Login failures
    - High port scans
    - Abnormal source IP behavior

Recommendations:

1. Enable or Tighten Rate Limiting: Limit login attempts to 3–5 per minute (if not already in place). This helps slow down brute-force attacks.
2. Enable Account Lockouts: Automatically lock user accounts after a number of failed login attempts.
3. Use Strong Password Policies: Enforce long, complex passwords that are hard to guess.
4. Change Default Credentials: Never leave default usernames or passwords unchanged.

Attack 2:

Nmap Scanning:

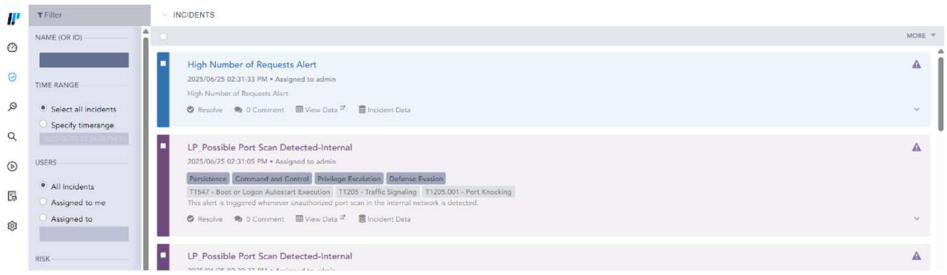
Nmap scanning is the process of using the Nmap (Network Mapper) tool to explore and gather information about networks and devices. It is widely used for network discovery, vulnerability assessment, and security auditing.

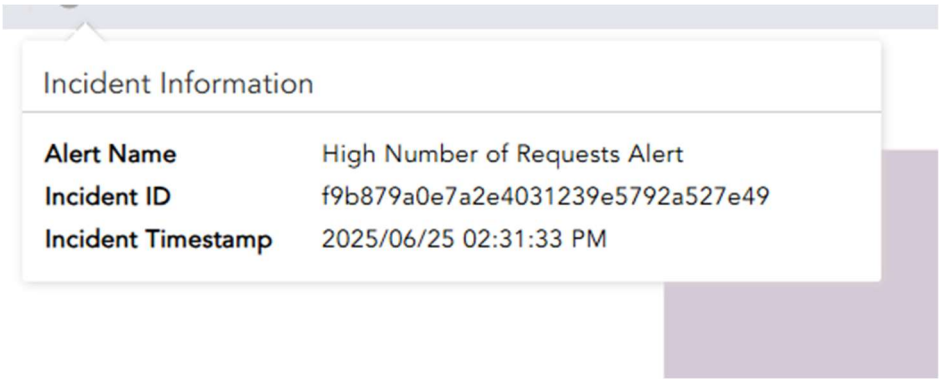
What was actually done?

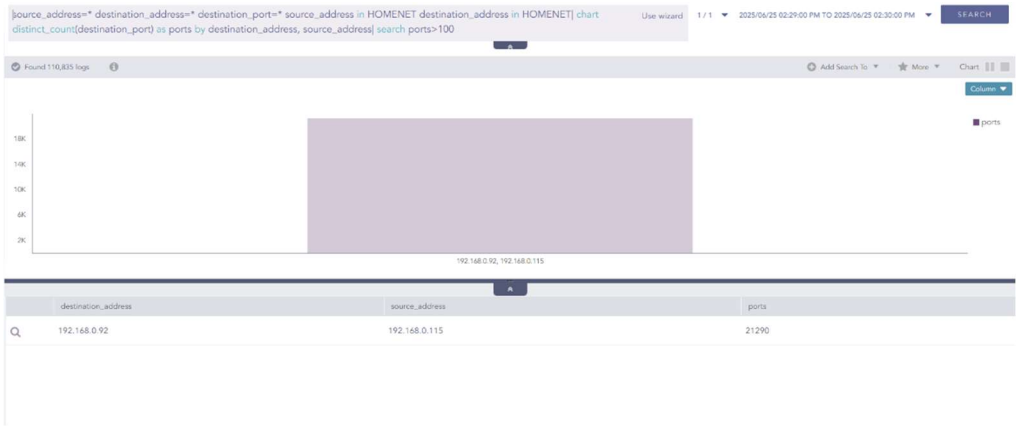
I performed nmap scan from the attacker kali linux to the victim machine. I performed a quick port scan using nmap 192.168.0.92 command. The command nmap scans a specific device on a network to discover open ports, running services, and other useful information about the target system. When executed, Nmap first checks if the device is online and then probes commonly used ports (like 22 for SSH or 80 for HTTP) to determine which ones are open and accessible.

What actually happened?

When nmap was performed, High Number of request Alert and LP_Possible Port Scan Detected-Internal (Default alert) was triggered. This alert shows that some device is trying to gain access to discover the open ports, running services, and other useful information about the target system.







What it means:

The High Number of Request alert indicates that a high volume of traffic — 135,288 logs — originated from the source IP 192.168.0.115 targeting the device 192.168.0.92 within a 5-minute window (from 02:24 PM to 02:29 PM). The search query filters logs by device IP and excludes internal traffic (excluding source IPs 192.168.0.92 and 192.168.0.9), then counts how many logs each unique external source IP generated. Since only one IP (192.168.0.115) surpassed the threshold of 500 logs, this suggests a potential scanning or brute force behavior — particularly suspicious given the volume and narrow timeframe.

The LP_Possible Port Scan Detected- Internal alert focuses on port scanning activity. It shows that the same source IP, 192.168.0.115, connected to 21,290 unique destination ports on the internal IP 192.168.0.92 within just one minute (02:29 PM to 02:30 PM). The search uses distinct_count(destination_port) to detect port sweeps and only reports results where more than 100 unique ports are accessed. This is a clear indication of a port scan, a reconnaissance technique often used by attackers to find open ports and services to exploit.

Immediate Action:

1. Immediately Block the Scanning IP Address
2. Rate Limit and Throttle Connections: Use firewall or router settings to limit:
    - Connection attempts per IP per second.
    - Number of new connections allowed in a short time (defeats Nmap speed).
3. Isolate the Target Host: Temporarily remove the target machine from the network.

Recommendation:

1. Harden Firewall Rules: "Deny all, allow specific" rule logic, Block unused ports completely.
2. Enable Port Knocking: Only open ports if a secret knock sequence is performed.
3. Use TCP Wrappers or iptables: Block IPs showing abnormal scanning behavior.
4. Implement Port Rate Limiting: Slow down connection attempts to many ports in a short time.